

Worked Example 3: Quantity Discounts, Cases 1-3

$A = 3.2, D = 2,400/\text{yr}, v_0 = 0.4, r = 0.24$

Inventory and Production Management

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Procedure (lecture slides 28-29)

$$v = v_0 \quad (Q < Q_1); \quad v = v_0(1 - d) \quad (Q \geq Q_1)$$

$$TRC(Q) = A\frac{D}{Q} + \frac{Q}{2}vr + Dv$$

Step 1

$$EOQ_{discount} = \sqrt{\frac{2AD}{v_0(1-d)r}}$$

Step 2

If $EOQ(d) \geq Q_1$, $EOQ(d)$ is the best order quantity (Case 3); otherwise go to step 3.

Step 3

$$TRC(EOQ) = \sqrt{2ADv_0r} + Dv_0 \quad \text{vs.} \quad TRC(Q_1)$$

If $TRC(EOQ) < TRC(Q_1)$ order the no-discount EOQ (Case 2); if $TRC(EOQ) > TRC(Q_1)$ order Q_1 (Case 1).

Offer: $d = 2.0\%$ for $Q \geq 500$

Solution trace

Step 1: EOQ at the discounted price $v_0(1-d) = 0.3920$: $EOQ(d) = \sqrt{2AD/(v_0(1-d)r)} = 404.06$

Step 2: $EOQ(d) = 404.06 < Q_1 = 500 \rightarrow$ go to step 3.

Step 3: EOQ without discount = $\sqrt{2AD/(v_0 r)} = 400.00$

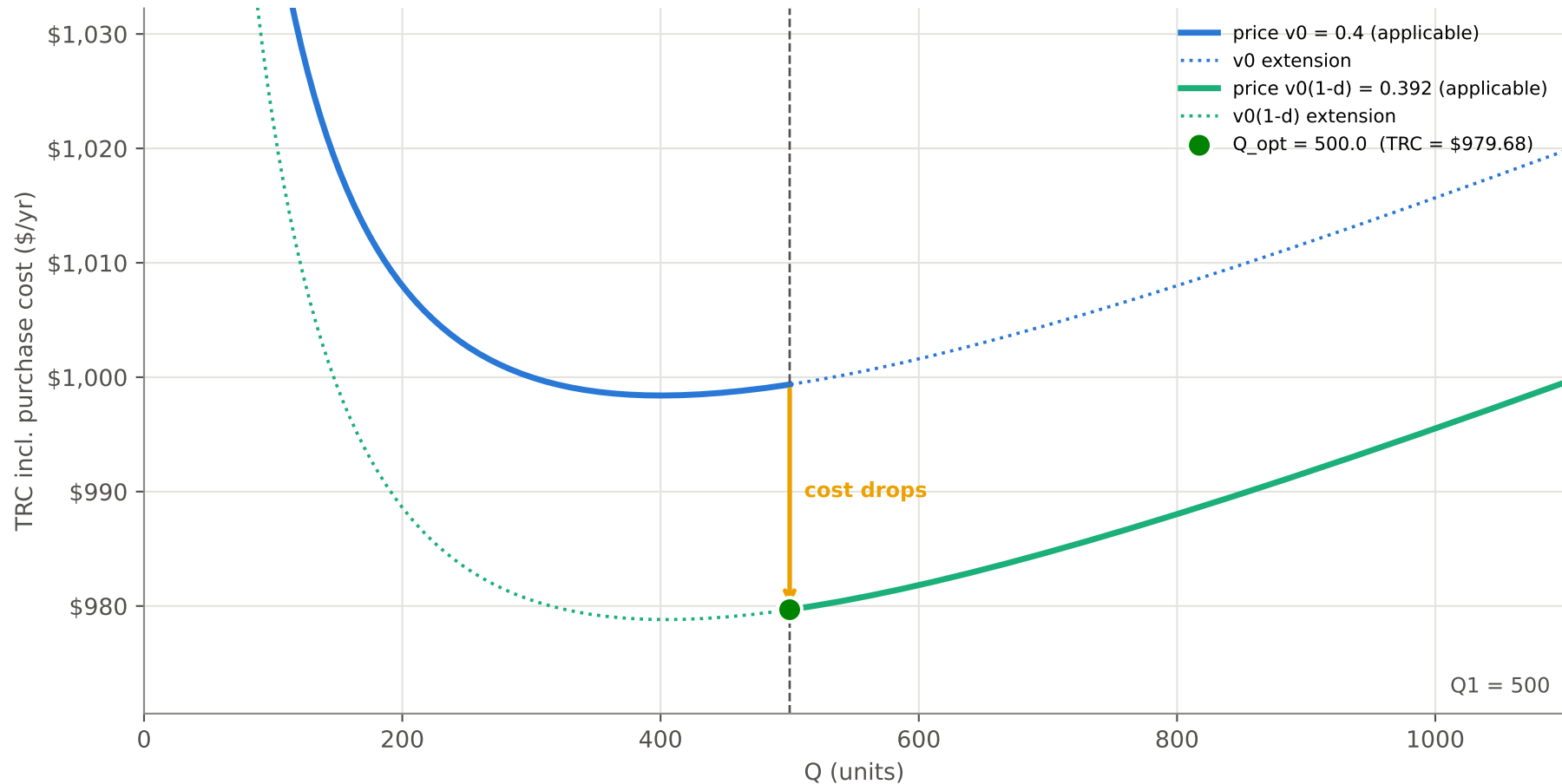
$$TRC(EOQ) = \sqrt{2AD v_0 r} + D v_0 = 998.40$$

$$TRC(Q_1) = AD/Q_1 + Q_1 v_0(1-d) r/2 + D v_0(1-d) = 979.68$$

$TRC(EOQ) > TRC(Q_1) \rightarrow$ take the discount, order exactly Q_1 (CASE 1).

RESULT: order $Q^* = 500.00$ units (Case 1); annual saving vs ignoring the offer = \$18.72

Case 1: $Q_1 = 500$, $d = 2\%$



Offer: $d = 1.0\%$ for $Q \geq 1200$

Solution trace

Step 1: EOQ at the discounted price $v_0(1-d) = 0.3960$: $EOQ(d) = \sqrt{2AD/(v_0(1-d)r)} = 402.02$

Step 2: $EOQ(d) = 402.02 < Q_1 = 1,200 \rightarrow$ go to step 3.

Step 3: EOQ without discount $= \sqrt{2AD/(v_0 r)} = 400.00$

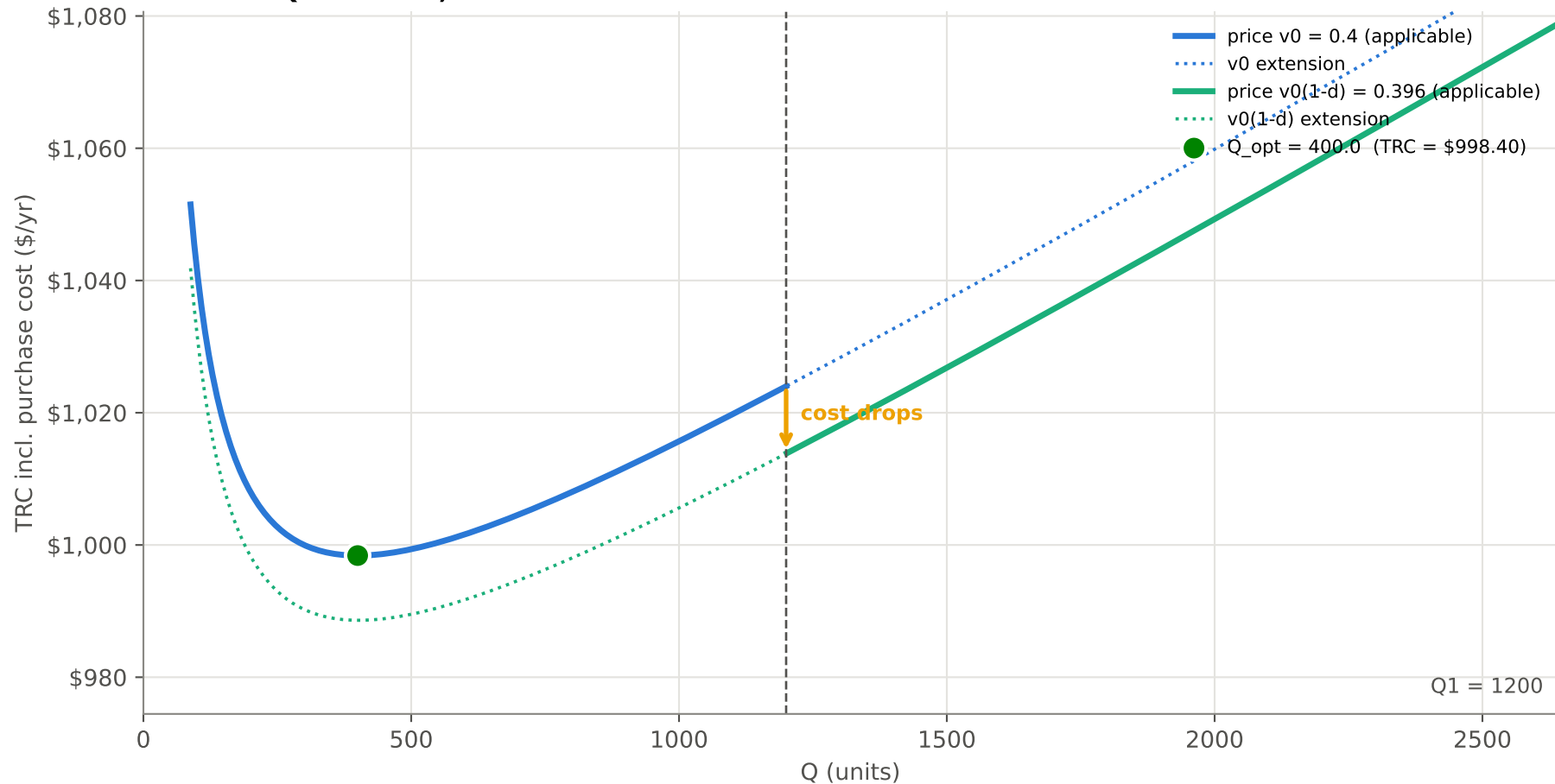
$$TRC(EOQ) = \sqrt{2AD v_0 r} + D v_0 = 998.40$$

$$TRC(Q_1) = AD/Q_1 + Q_1 v_0(1-d) r/2 + D v_0(1-d) = 1,013.82$$

$$TRC(EOQ) < TRC(Q_1) \rightarrow \text{ignore the discount, order EOQ (CASE 2).}$$

RESULT: order $Q^* = 400.00$ units (Case 2); annual saving vs ignoring the offer = \$0.00

Case 2: $Q_1 = 1200$, $d = 1\%$



Offer: $d = 2.0\%$ for $Q \geq 300$

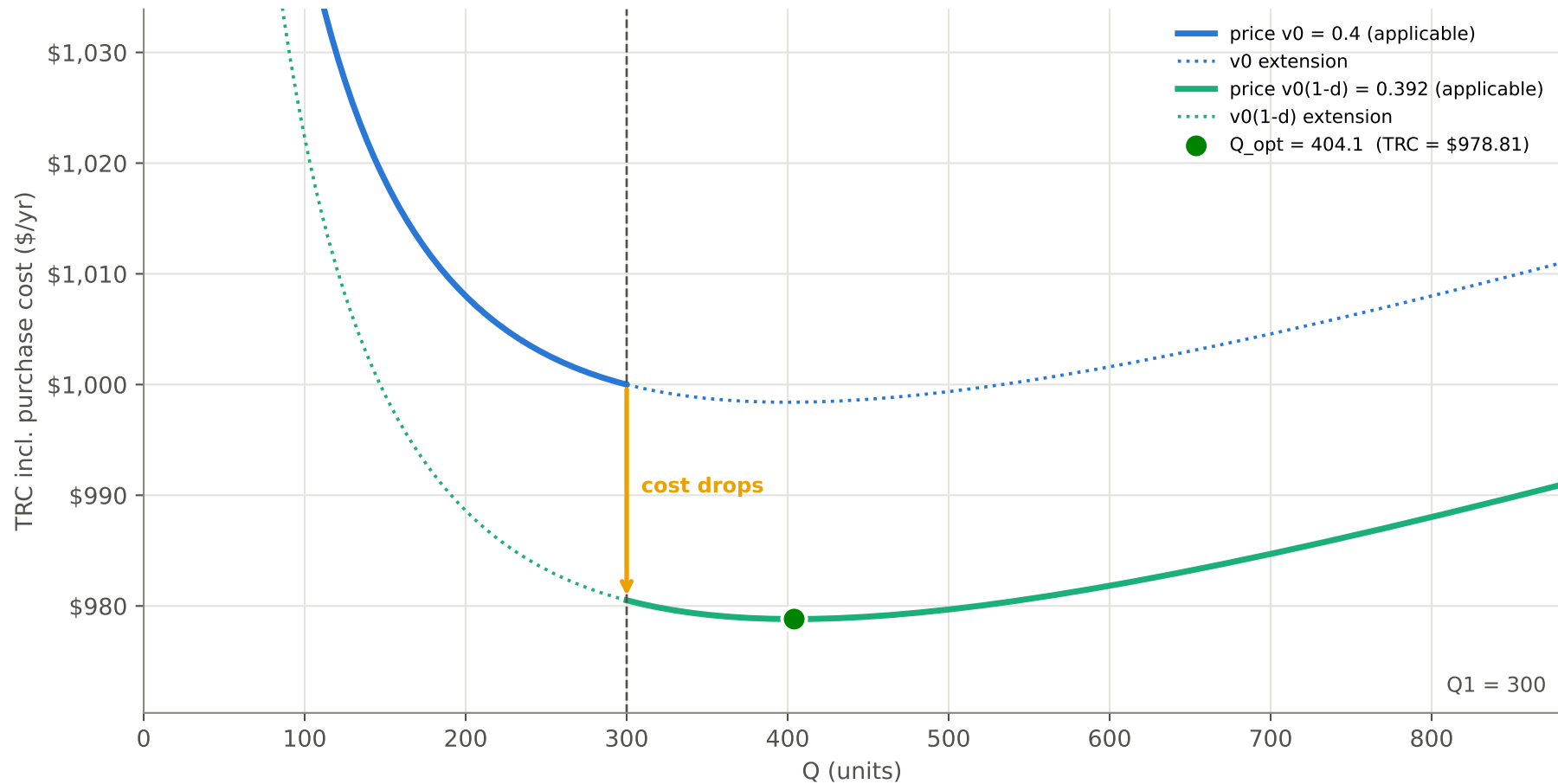
Solution trace

Step 1: EOQ at the discounted price $v_0(1-d) = 0.3920$: $EOQ(d) = \sqrt{2AD/(v_0(1-d)r)} = 404.06$

Step 2: $EOQ(d) = 404.06 \geq Q_1 = 300 \rightarrow EOQ(d)$ is feasible and optimal (CASE 3). Stop.

RESULT: order $Q^* = 404.06$ units (Case 3); annual saving vs ignoring the offer = \$19.59

Case 3: $Q_1 = 300$, $d = 2\%$



Summary of all offers

d	Q1	EOQ(d)	EOQ(0)	TRC(EOQ0)	TRC(Q1)	Case	Q*	Saving \$/yr
2.0%	500.00	404.06	400.00	998.40	979.68	1.00	500.00	18.72
1.0%	1,200.00	402.02	400.00	998.40	1,013.82	2.00	400.00	0.00
2.0%	300.00	404.06	400.00	998.40	980.51	3.00	404.06	19.59

Discount decision map (A=3.2, D=2,400, v0=0.4, r=0.24)

